

IoT Lab 3

IEEE 802.15.4 and CoAP

Client-Server Communication · Zephyr RTOS on nRF52840 · Execution on real Hardware

Part 1 – UDP Setup

Part 2 – CoAP Server

Part 3 – CoAP Client

Part 4 – Evaluation

Part 1 – UDP Setup

Configuration

- IEEE 802.15.4 PAN ID = 0x1234
- Channel = 15
- IPv6 addresses fd00::1 / fd00::2
- UDP Port 12345

Communication Flow

- Initialize network interface
- Create UDP socket
- Sender sends temp values
- Receiver receives and prints

Sender

```
// Send data to receiver
uint16_t temp = 20;
while (1) {
    char msg[10];
    snprintf(msg, sizeof(msg), "temp=%d", temp);
    ret = sendto(sock, msg, strlen(msg), 0, (struct sockaddr *)&dest_addr,
                sizeof(dest_addr));
    if (ret < 0) {
        printk("send() failed: %d\n", errno);
    } else {
        printk("Sent: %s\n", msg);
    }
    k_sleep(K_SECONDS(2));
    temp ++;
};
```

Receiver

```
// Receive data from sender
while (1) {
    char buffer[10];
    struct sockaddr_in6 src_addr;
    socklen_t src_len = sizeof(src_addr);
    printk("Waiting for data...\n");
    ret = recvfrom(sock, buffer, sizeof(buffer) - 1, 0,
                  (struct sockaddr *)&src_addr, &src_len);
    if (ret < 0) {
        printk("recv() failed: %d\n", errno);
        continue;
    }
    printk("Received: %s\n", buffer);
}
```

Part 2 – CoAP Server

Endpoints

1

GET /sensor

Returns simulated temperature

2

POST /action

Toggles action state (True/False)

Server Workflow

1. Receive UDP Datagram from Client IP Address
2. Parse CoAP Packet
3. Match URI
4. Generate ACK Response + Payload

Part 3 – CoAP Client

1

Periodic Requests

Get /sensor every 5s
Receive ACK
Print Temperature
Measure RTT

2

Button Trigger

GPIO Interrupt
POST /action
Server toggles state
Confirmation received

3

Use Polling

2s Timeout
=> Re-Send Request if no response received

4

Check Message-ID

Identify incoming ACKs via message_id to match corresponding requests

```
// Send CoAP request to server
printk("Sending CoAP request to server...\n");
printk("REQ MID=%u\n", msg_id);
int64_t start_time = k_uptime_get();
ret = sendto(sock,
    request.data,
    request.offset,
    0,
    (struct sockaddr *)&server_addr,
    sizeof(server_addr));
if (ret < 0) {
    printk("sendto response failed: %d\n", errno);
    k_sleep(K_SECONDS(5));
    continue;
}
printk("sendto successful, bytes=%d\n", ret);
```

```
if (fds[0].revents & POLLIN) {
    ret = recvfrom(sock,
        recv_buf,
        sizeof(recv_buf),
        0,
        (struct sockaddr *)&server_response_addr,
        &server_len);

    if (ret < 0) {
        printk("recvfrom() failed: %d\n", errno);
        k_sleep(K_SECONDS(5));
        continue;
    }
}
```

```
// Parse CoAP response
ret = coap_packet_parse(&response, recv_buf, ret, NULL, 0);
if (ret < 0) {
    printk("Invalid CoAP packet: %d\n", ret);
    continue;
}

uint8_t conn_type = coap_header_get_type(&response);
uint16_t id = coap_header_get_id(&response);
uint8_t rx_token_len;
const uint8_t *rx_token = coap_header_get_token(&response, &rx_token_len);
// Check if response matches the request
if (conn_type == COAP_TYPE_ACK && id == msg_id) { //&& rx_token_len == size
//memcmp(rx_token, token, sizeof(token)) == 0) { => token parsing does not
// so we just check message ID for matching
    int64_t end_time = k_uptime_get();
    printk("Received ACK for message ID: %d\n", id);
}
```

Part 4 – Evaluation Results

Executed on Real Hardware - Because of Laptop USB Connection Devices were close to each other

100%

Reliability

all nodes successfully forward
all messages

**~13 - 15
ms**

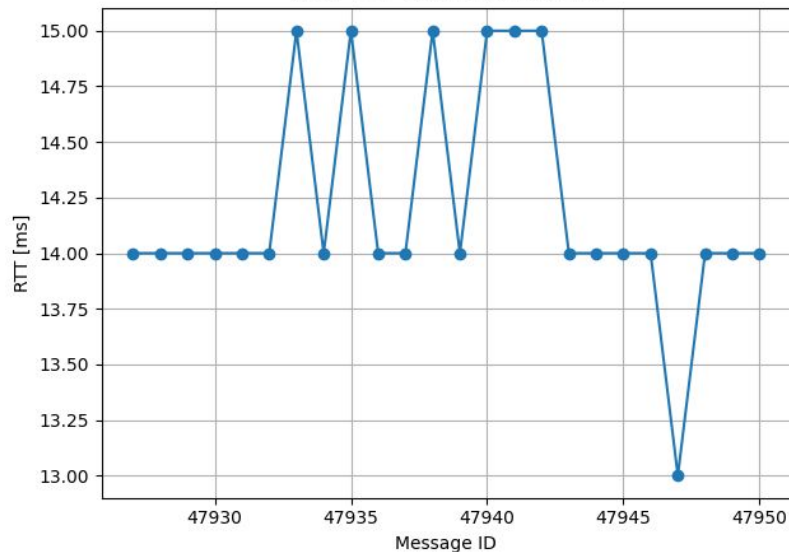
Round-Trip-Time

Average 14 ms RTT

CoAP reliability - normal



CoAP RTT over time - normal



Discussion

Advantages

- Reliable CoAP communication
- Low RTT
- Simple REST-like API
- IPv6-based networking

Limitations

- Only two nodes
- No interference tests
- BLE comparison only conceptual.

Summary

Achievements

- IEEE 802.15.4 configured
- UDP Communication
- CoAP Server
- CoAP Client
- Performance evaluation

Key Takeaways

- **High Reliability**
- **Low RTT**
- **Responsive Button Control**
- **Suitable for IoT Application**